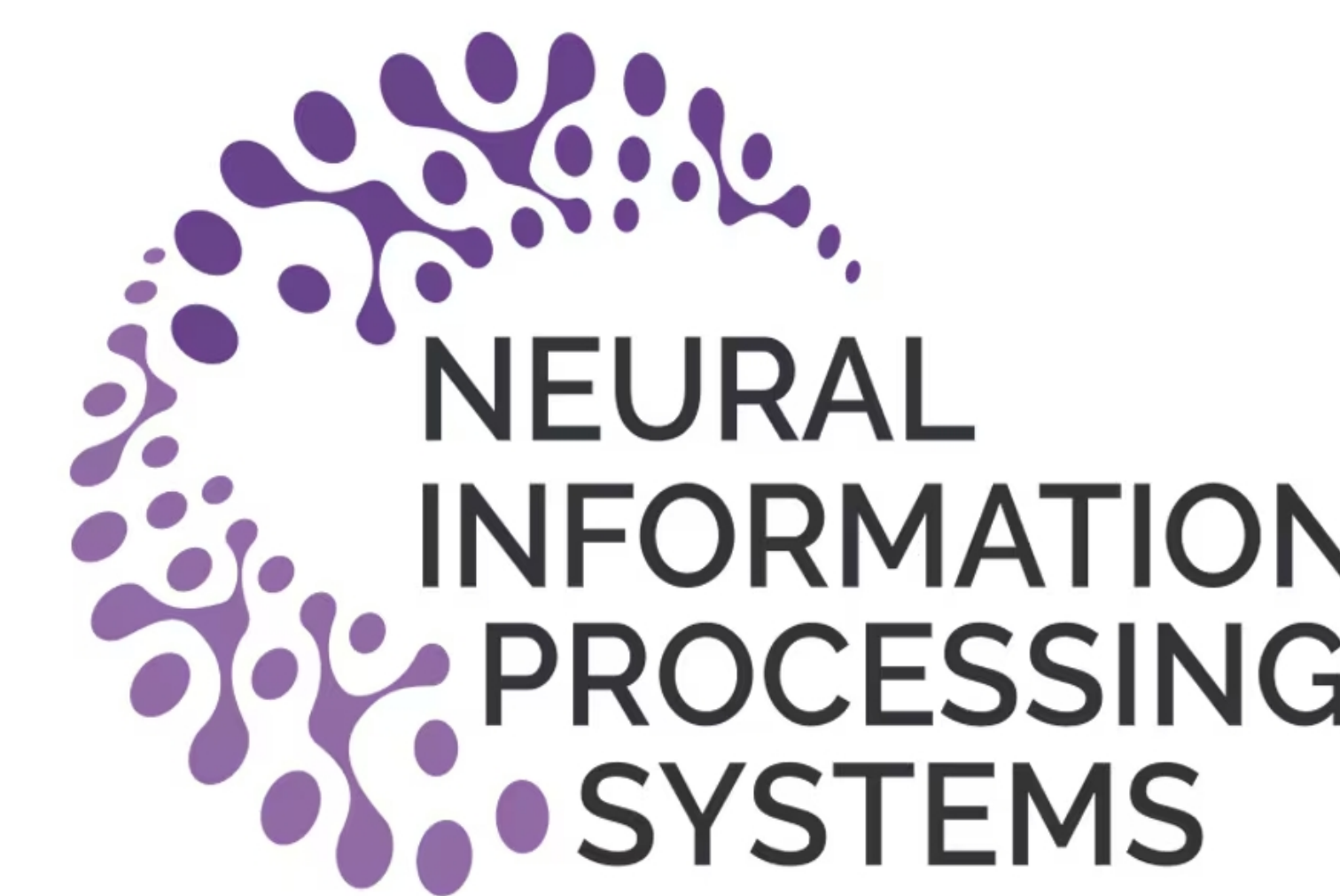




BReD Block Replay Dithering for Stable Low-Bit EMA Optimizer States

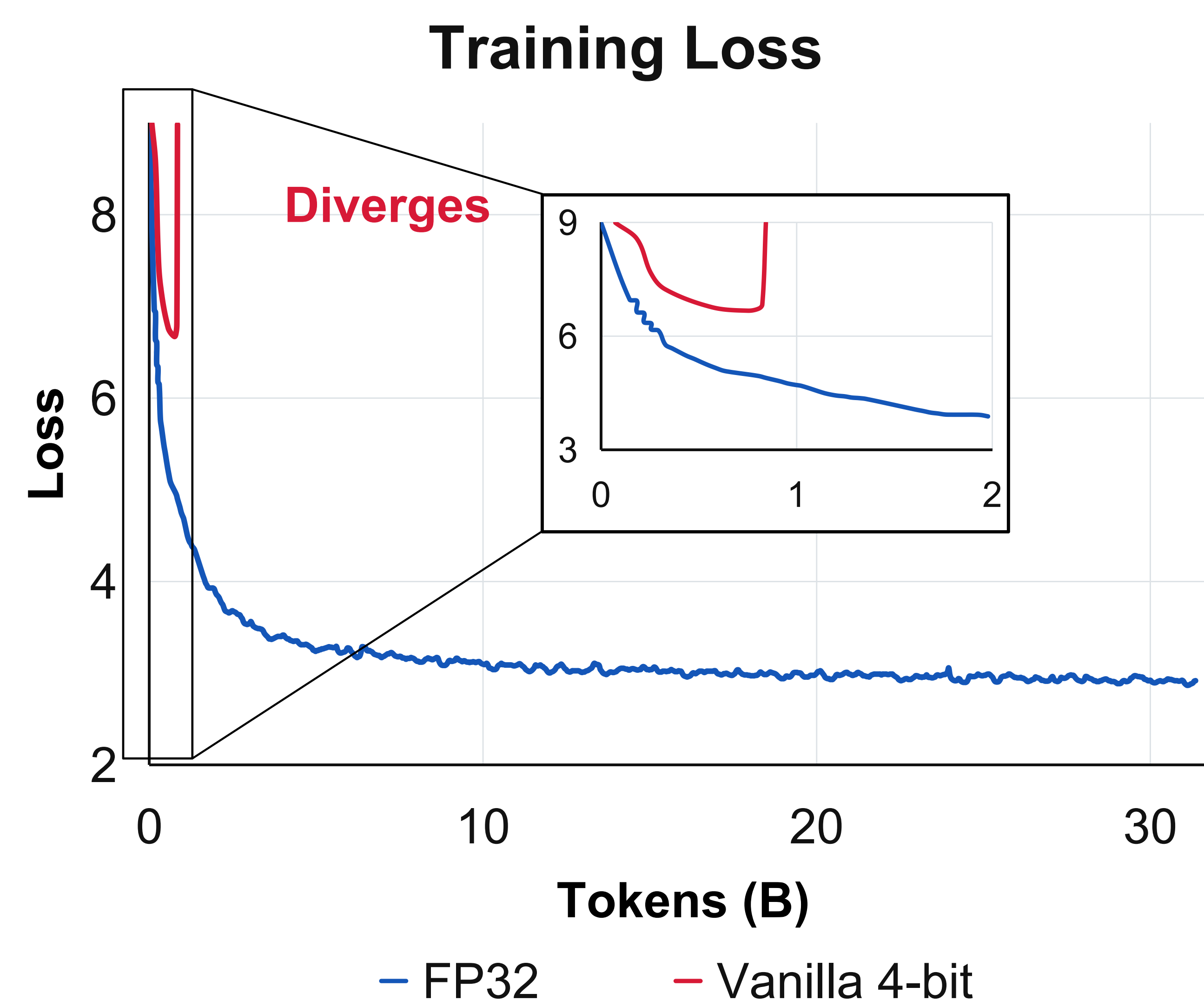


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1 Quantize optimizer state can diverge

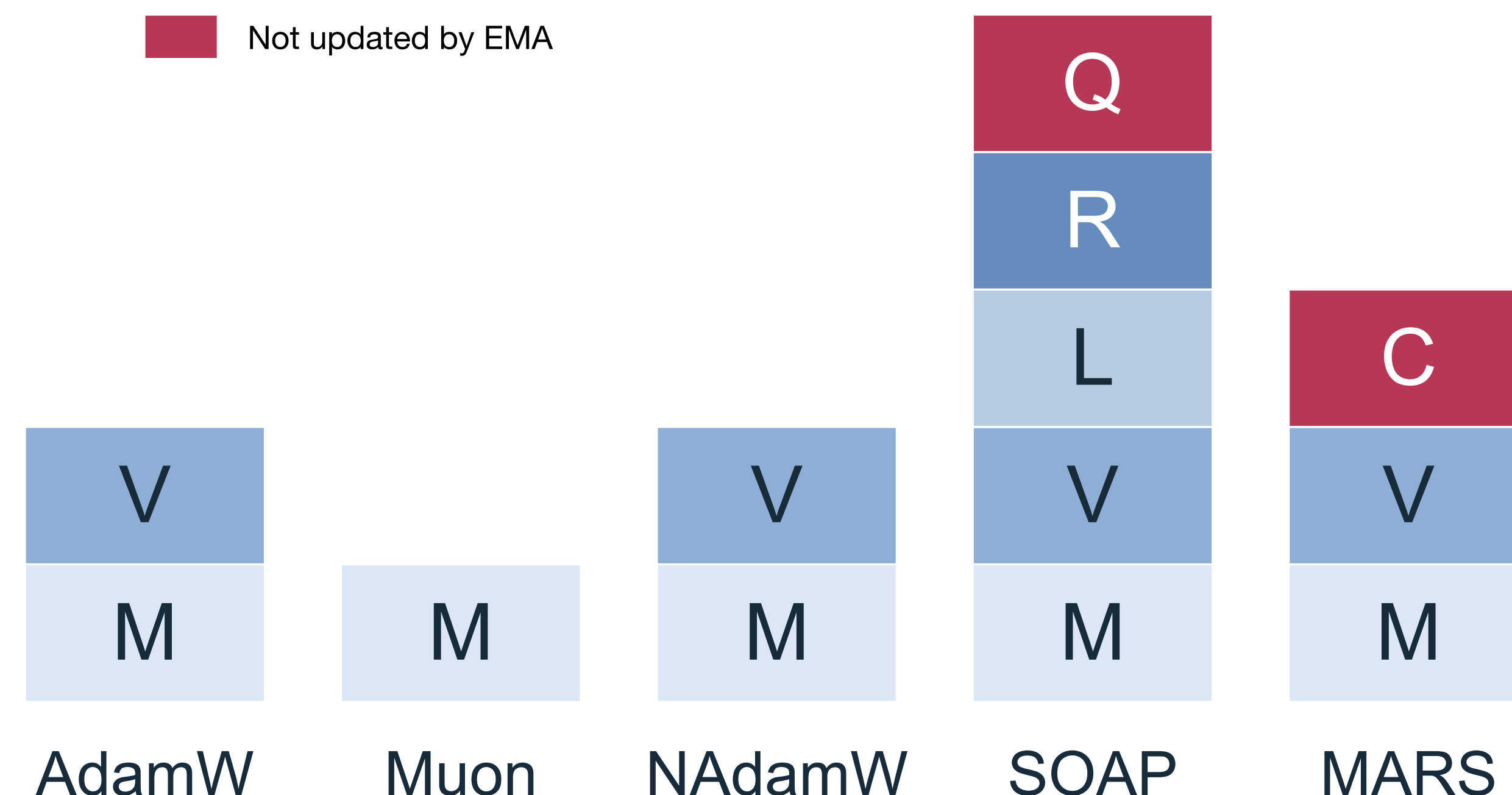


Train by Adam, quantize 1st and 2nd moment.

Optimizer states require extra memory

Each block $\approx$ one model-weight tensor in memory

Updated by EMA
Not updated by EMA



2 What goes wrong?

Exponential moving average (EMA)

$$s_{t+1} = \beta s_t + (1 - \beta) u_{t+1}$$

s: state u: incoming update β : decay

$$e_{t+1} = \beta e_t + \varepsilon_{t+1}$$

State error = retained error + new rounding error

State error builds up during training



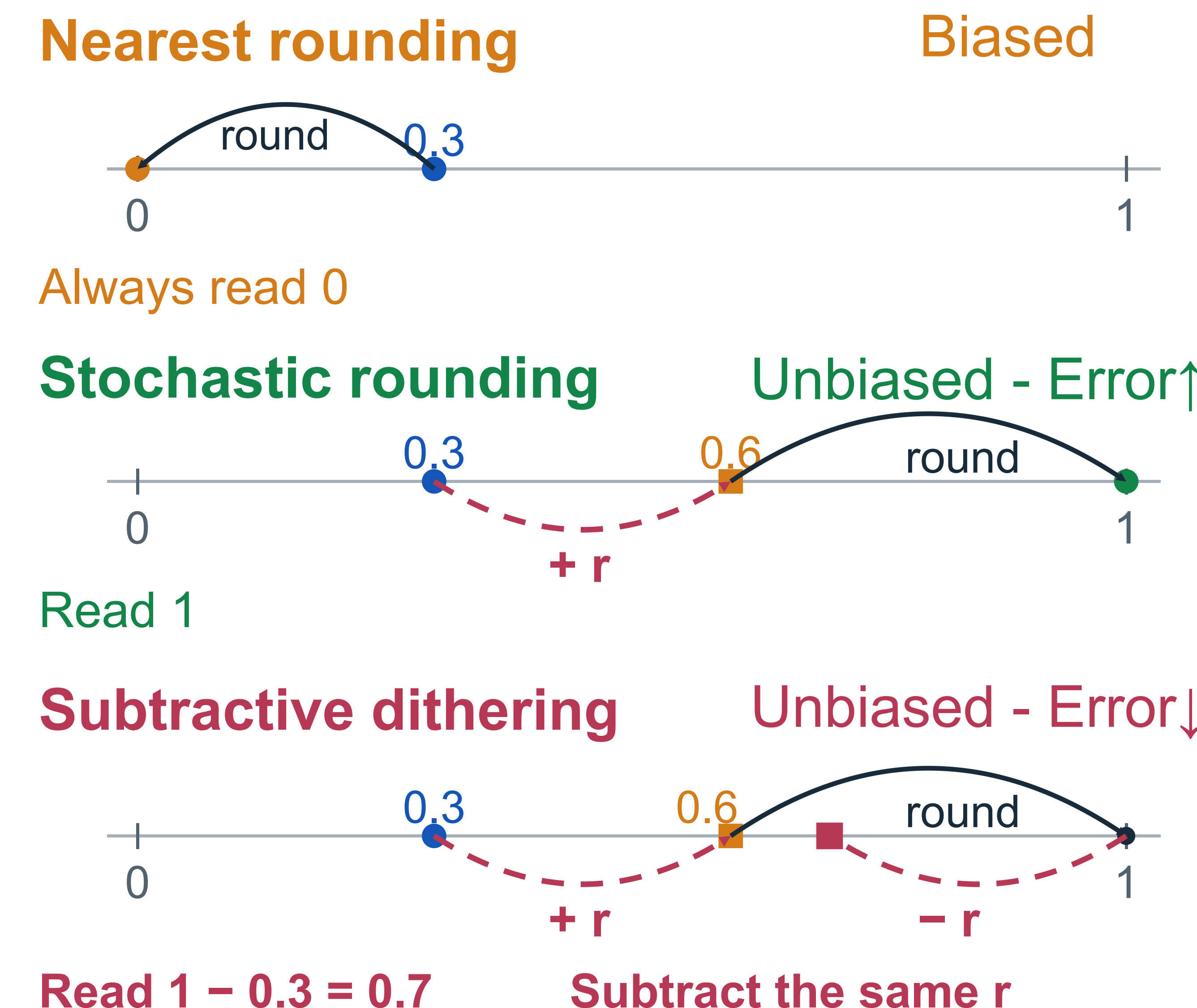
Theorem 3.2 (Steady-state second moment under quantized EMA). Assume $\{\epsilon_t\}_{t \geq 1}$ are i.i.d. with $\mathbb{E}[\epsilon_t] = b$ and $\mathbb{E}\|\epsilon_t - b\|_F^2 = \sigma^2$. For $|\beta| < 1$, the state error satisfies

$$\lim_{t \rightarrow \infty} \mathbb{E}\|e_t\|_F^2 = \underbrace{\frac{\|b\|_F^2}{(1 - \beta)^2}}_{\text{mean (bias) term}} + \underbrace{\frac{\sigma^2}{1 - \beta^2}}_{\text{variance (noise) term}}. \quad (9)$$

Error accumulated over time.
 β controls how strongly the errors amplified.

3 BReD: low-bit quantization

Same input: $x = 0.3$



Example draw: $r = +0.3$; unit spacing, no clipping.

BReD replays a random shift per block

Controlled 4-bit comparison

120M pretraining · Validation PPL $\downarrow$

Optimizer	FP32 state	NR	SR	BReD
AdamW	18.3	Diverged	Diverged	Stable (18.7)
Muon	17.8	Degraded (31.0)	Degraded (21.7)	Stable (18.0)
NAdamW	18.9	Diverged	Diverged	Stable (18.9)
SOAP	17.6	Diverged	Diverged	Stable (18.1)
MARS	18.7	Diverged	Diverged	Stable (18.9)